

AKSENOVO-ZILOVSKOE, Russian Federation

WMO: 306690

Lat: 53.067N

Lon: 117.483E

Elev: 2323

StdP: 13.50

Time Zone: 9.00 (E09)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.9	-26.2	-35.9	0.8	-30.4	-31.4	1.0	-25.3	12.8	-0.8	10.8	-5.8	1.8	240	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	81.0	62.2	77.7	61.2	74.8	60.9	66.4	75.1	64.7	73.4	63.1	71.2	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	95.8	69.8	61.8	89.9	68.1	60.0	84.2	66.9	32.5	75.0	31.2	73.5	29.9	71.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.8	9.9	-34.4	87.6	5.9	2.6	-38.6	89.5	-42.1	91.0	-45.4	92.5	-49.6	94.4
			-33.9	70.3	5.8	2.7	-38.1	72.2	-41.5	73.8	-44.8	75.4	-49.0	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	-9.0	-0.1	13.1	31.9	46.5	58.2	62.0	58.8	45.7	29.7	7.0	-5.8	(6)
	DBStd	26.51	8.36	7.57	11.10	8.64	8.43	6.18	4.66	4.83	6.83	8.20	10.21	8.29	(7)
	HDD50	8917	1829	1403	1144	545	168	10	1	2	164	630	1290	1731	(8)
	HDD65	13429	2294	1823	1609	994	574	214	113	199	578	1095	1740	2196	(9)
	CDD50	1000	0	0	0	2	60	255	372	275	36	0	0	0	(10)
	CDD65	35	0	0	0	0	1	8	19	7	0	0	0	0	(11)
	CDH74	813	0	0	0	5	92	272	308	134	2	0	0	0	(12)
	CDH80	133	0	0	0	1	10	45	61	16	0	0	0	0	(13)
	WSAvg	3.4	3.1	3.0	3.7	4.6	4.5	3.4	3.0	2.7	3.2	3.3	3.2	3.1	(14)
(15) Precipitation	PrecAvg	17.1	0.2	0.2	0.2	0.8	1.5	2.9	4.0	3.5	2.2	0.7	0.5	0.3	(15)
	PrecMax	23.5	0.6	0.6	0.4	2.1	3.0	5.2	8.9	5.7	3.9	1.8	1.2	0.8	(16)
	PrecMin	12.9	0.0	0.0	0.0	0.2	0.2	1.1	1.2	2.2	0.6	0.0	0.1	0.1	(17)
	PrecStd	2.4	0.1	0.1	0.1	0.5	0.7	1.1	1.7	0.9	0.9	0.4	0.2	0.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	25.2	51.0	68.2	81.1	84.4	86.2	82.4	73.6	59.6	41.3	21.7	(19)
		MCWB	13.6	20.6	38.8	48.8	58.5	64.8	64.6	65.6	57.1	46.0	34.0	19.2	(20)
	2%	DB	11.0	21.2	42.8	59.2	76.5	80.3	80.7	77.7	69.5	53.3	32.5	14.6	(21)
		MCWB	9.1	17.6	34.0	44.4	57.0	61.0	63.8	63.4	55.2	41.9	27.9	12.1	(22)
	5%	DB	7.2	18.3	37.2	52.4	70.8	77.0	77.0	74.2	64.2	49.0	27.2	10.1	(23)
		MCWB	5.6	15.3	29.9	40.3	52.9	59.9	62.6	61.8	53.0	39.4	23.6	8.2	(24)
	10%	DB	3.7	15.0	32.3	46.8	64.8	73.3	74.2	70.9	59.7	43.8	22.4	5.9	(25)
		MCWB	2.5	12.4	26.3	36.8	49.5	59.2	62.4	60.2	49.9	35.7	19.4	4.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	21.1	39.5	49.5	60.3	67.4	68.4	68.3	59.9	46.0	33.7	19.4	(27)
		MCDB	16.4	25.0	50.8	65.8	78.5	80.3	76.1	77.5	67.9	58.8	40.4	21.2	(28)
	2%	WB	9.1	17.7	34.0	44.5	57.0	64.3	66.7	65.7	56.9	42.4	28.5	12.4	(29)
		MCDB	10.8	20.8	42.4	58.7	74.3	75.5	74.9	73.9	64.5	52.0	32.1	14.6	(30)
	5%	WB	5.7	15.3	30.0	40.6	53.5	62.6	65.4	63.5	54.8	39.5	23.6	8.3	(31)
		MCDB	7.1	18.2	36.8	51.4	69.5	72.6	73.8	70.8	62.4	48.6	27.2	10.0	(32)
	10%	WB	2.6	12.5	26.6	37.4	50.5	60.6	63.9	61.5	51.5	36.1	19.5	4.6	(33)
		MCDB	3.7	15.0	32.1	44.9	63.9	69.2	71.9	68.3	59.6	43.1	22.4	5.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.2	25.4	25.8	22.4	25.2	23.3	21.4	21.1	22.1	22.1	20.8	17.5	(35)
		MCDBR	22.6	28.0	27.8	32.9	36.2	32.3	29.1	28.4	31.2	27.8	22.9	20.6	(36)
	5% WB	MCWBR	20.6	24.8	20.1	20.4	19.2	15.8	14.9	15.0	17.9	17.9	18.3	18.5	(37)
		MCWBR	22.2	27.9	26.4	30.9	34.6	24.9	22.7	22.9	26.1	27.3	23.0	20.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.246	0.274	0.318	0.423	0.532	0.484	0.428	0.397	0.352	0.307	0.272	0.252	(40)
		taud	2.194	2.167	2.131	1.989	1.776	2.013	2.244	2.309	2.377	2.394	2.255	2.133	(41)
	Ebn at Noon	Ebn at Noon	236	263	272	253	230	246	259	261	261	250	222	204	(42)
		Edh at Noon	20	29	37	50	65	52	41	36	30	23	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	416	782	1319	1658	1763	1892	1726	1485	1146	774	449	313	(44)
	RadStd	RadStd	10	14	36	97	127	134	130	87	86	54	23	10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page